3H-PSNet: Hierarchical PointSAM with contrastive refinement and risk-aware distillation for 3D semantic segmentation

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Agenda

- Introduction
- Literature Review
- Research Gaps Identified
- Problem Statement
- Objectives
- Proposed Methodology
- Implementation Details
- Results
- Conclusion
- References

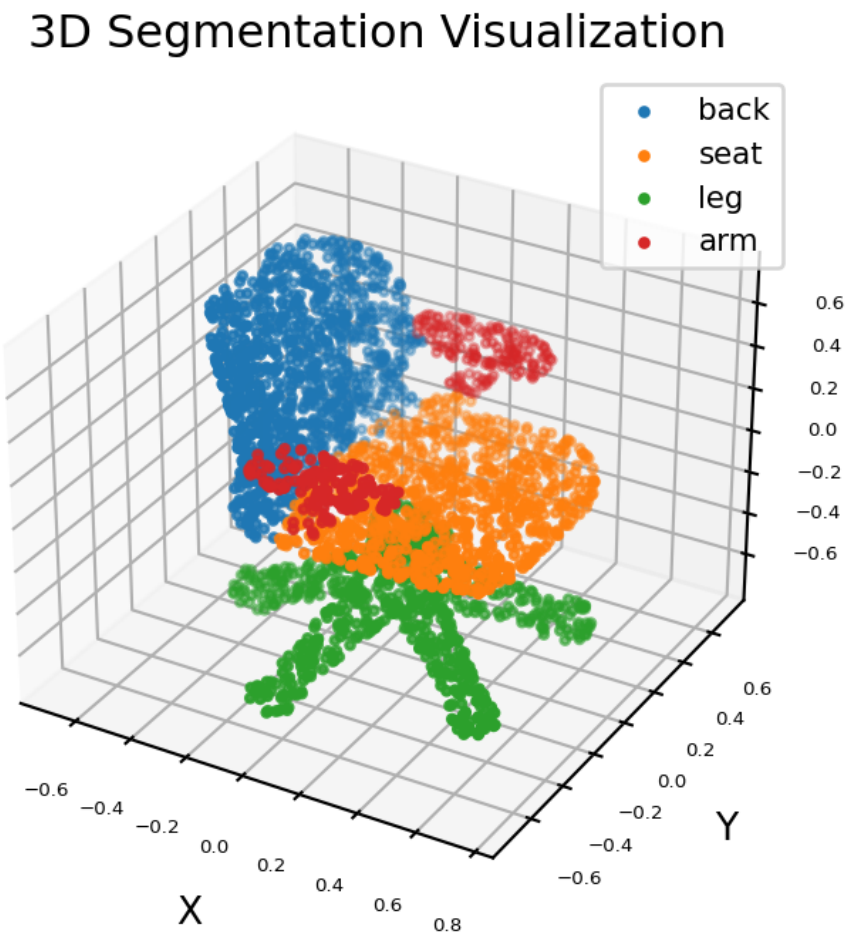
Introduction: Point Clouds and Semantic Segmentation

1. **What are Point Clouds?**
   - A point cloud is a collection of discrete data points in a 3D coordinate system, defined as $P = \{p_i\}_{i=1}^{N} \in \mathbb{R}^N$
   - Every point can be represented by its features: $x_i =< F_{1i}, F_{2i}, \ldots, F_{Ni} >$, where F_{pi} represents the pth feature (e.g., spatial coordinates or color)

2. **Definition of Semantic Segmentation**
   - Semantic segmentation is the task of assigning a specific class label y_i from a class vector $Y =< y_1, y_2, \ldots, y_m >$ to every point in the dataset X
   - In part segmentation, the goal is to predict a binary mask $M \in \{0, 1\}^N$ for a specific component (e.g., the leg of a chair) given a prompt coordinate

3. Example of Semantic Segmentation

- **Example:** Segmenting a 3D scene using PointNet where the model distinguishes between "Chair legs," "Seats," 'arms," and "Back"



Introduction: Challenges and Motivation

Challenges in Skewed Training Sets

- Skewed distributions lead to feature-space dominance by frequent categories, where majority classes dictate the global geometric representation
- **Gradient Bias:** Training optimization is skewed toward reducing global error, effectively suppressing gradients from rare, underrepresented classes
- **Decision Boundary Shift:** Networks develop biased boundaries that "subsume" minority regions into neighboring majority clusters

Rare Class Inefficiency

- Minority classes suffer from weak representation learning, resulting in fragmented or unstable segmentation for sparse and irregular structures
- **Inertial Learning:** Standard backpropagation fails to distill unique geometric features for rare parts, leading to semantic confusion in spatially adjacent regions
- Traditional class-agnostic sampling reinforces these inefficiencies by failing to prioritize high-risk, low-relief boundary points

Impact: Hierarchical PointSAM with Contrastive Refinement and Risk-aware Distillation

- Poor quality in rare categories hinders precision-critical applications in robotics, AR/VR, and autonomous scene understanding
- **Contrastive Refinement:** Subtractive prompting is required to explicitly push decision boundaries away from confusion zones between rare and frequent parts
- **Risk-aware Distillation:** Learning to distill geometric complexity allows the model to prioritize high-risk regions that are otherwise overlooked by standard segmentation losses

Literature Review

Paper	Year and Venue	Key Points	Gaps	Dataset	Model
PointSAM: Pointly-Supervised Segment Anything Model for Remote Sensing Images	2025, IEEE Transactions on Geoscience and Remote Sensing (IEEE TGRS)	Adapts Segment Anything Model(SAM) to operate with sparse point-level supervision instead of dense masks. Significantly reduces annotation cost while maintaining strong segmentation accuracy and generalization across remote sensing imagery.	No adaptive class imbance handling mechanism beyond hungarian matching.	NWPU VHR-10, WHU, and HRSID	SAM (ViT backbone) with point prompts

Literature Review

Paper	Year and Venue	Key Points	Gaps	Dataset	Model
SAMNet++: A Segment Anything Model for Supervised 3D Point Cloud Semantic Segmentation	2025, Remote Sensing (MDPI)	Introduces a hybrid 2D–3D pipeline combining SAM-based mask generation with PointNet++ refinement. Achieves competitive segmentation performance with lower computational overhead than fully 3D point-cloud networks	for large-scale outdoor environments where the transition from unsupervised 2D priors to 3D geometric refinement often results in contextual information loss	UAV LiDAR, public 3D benchmarks	SAM-based masks + PointNet++ refinement

Literature Review

Paper	Year and Venue	Key Points	Gaps	Dataset	Model
Generating Inverse Feature Space for Class Imbalance in Point Cloud Semantic Segmentation	2025 / IEEE TPAMI	- Proposes **InvSpaceNet**: dual-branch network using an inverse feature space with momentum-updated prototypes and contrastive loss. - Instance-balanced + inverse sampling branches correct minority-class bias. - Dynamic re-weighting improves imbalance learning. - Strong gains across four datasets, especially minority classes.	- Classical imbalance solutions (weighted loss, brute sampling) don't generalize well to segmentation. - Extreme imbalance still remains difficult. - Open challenge: strengthening minority-class representation under severe imbalance.	S3DIS; ScanNet v2; Toronto-3D; Semantic KITTI	Dual-branch design; backbones: MLP (PointNet++), CNN (SPVCNN, MinkowskiNet), Transformer (PTv2), RNN (RSNet).

Literature Review

Paper	Year and Venue	Key Points	Gaps	Dataset	Model
3D-AOCL: Analytic online continual learning for imbalanced 3D point cloud classification	2024 / Science Direct	- Focuses on class imbalance for online continual point-cloud classification. - Distance-Based Sampling (DBS) for feature-space oversampling; Balanced Analytical Learning (BAL) for dynamic classifier weighting. - Feature fusion for V2X vehicle/infrastructure streams. - Reports higher mean class accuracy with lower parameter cost than prior CL methods.	- Existing CL replay/regularization for images not tailored to point clouds. - BAL requires careful manual tuning; unstable if mis-set. - Focused on classification (not segmentation); evaluated on single-stream scenarios.	V2X-Seq; ModelNet40 (pre-training)	PointNet backbone; MLP analytic classifier; DBS oversamples minority features; BAL = dynamic analytic weights.

Literature Review

Paper	Year and Venue	Key Points	Gaps	Dataset	Model
Fairness-Aware Boosting model for Imbalanced 3D Point Cloud Segmentation in Autonomous Driving	2025 / IEEE CVPR	- Adds AdaBoost ensemble to RandLA-Net to target misclassified/rare classes. - Introduces Weighted IoU (wIoU) for fairer evaluation. - Improves minority classes significantly.	- Confusion persists among similar classes. - Ensemble needs more tuning + cross-dataset validation. - Class-similarity issues remain.	Toronto-3D; Semantic3D	RandLA-Net + AdaBoost; wIoU metric

Literature Review

Paper	**Year and Venue**	**Key Points**	**Gaps**	**Dataset**	**Model**
Subclassified Loss: Rethinking Data Imbalance From Subclass Perspective for Semantic Segmentation	2024 / IEEE TIV	- Proposes *subclassified loss* that discovers virtual subclasses via pixel similarity and reweights loss per-subclass. - Compatible with multiple segmentation backbones; improves mIoU for minority/complex subclasses.	- Gains vary; limited improvement where boundary overlap is strong. - Subclass discovery sensitive to similarity metric choice.	Semantic KITTI; Cityscapes	RangeNet, KPRNet, PointRend, STDC, SegFormer.

Literature Review

Paper	Year and Venue	Key Points	Gaps	Dataset	Model
Effective Class-Imbalance Learning Based on SMOTE and Convolutional Neural Networks	2023 / Applied Sciences	- Combines SMOTE oversampling + normalization + CNN (SMOTE-NORM-CNN) for binary imbalanced datasets. - Benchmarks resampling strategies (SMOTE, TL, OSS, NearMiss) with DNN/CNN; SMOTE-NORM-CNN often best for accuracy / F1 / recall. - Thorough evaluation on 24 datasets.	- Training cost/time higher for CNNs vs classical ML. - SMOTE can introduce noise / distort boundaries; no multi-class evaluation in study. - Generalization to multi-class / time-series unclear.	KEEL (20+ datasets); Breast Cancer; Z-Alizadeh Sani	CNN (2 conv + 2 dense, ReLU, focal loss); DNN.

Literature Review

Paper	Year and Venue	Key Points	Gaps	Dataset	Model
Addressing the class imbalance problem in network intrusion detection systems using data resampling and deep learning	2023 / The Journal of Supercomputing	- Uses ADASYN (oversampling) + Tomek Links (undersampling) preprocessing; tested with several deep learning architectures. - Proposes two-stage IDS (detect attack, then classify type) to improve minority detection.	- Risk of overfitting due to synthetic samples; preprocessing choices are critical. - Tested only on NSL-KDD; broader applicability needs validation.	NSL-KDD	MLP, DNN, CNN, CNN-BLSTM; ADASYN + Tomek Links.

Literature Review

Paper	Year and Venue	Key Points	Gaps	Dataset	Model
Boosting Minority Class Prediction on Imbalanced Point Cloud Data	2020 / MDPI	- Two-stage framework: Hybrid-Sampling + Balanced Focus Loss. - Reports PointNet accuracy improvements (6–7%).	- Oversampling may cause overfitting; undersampling may drop useful data; weighted losses need tuning.	Custom point cloud (Orbbec Astra S, 480×640; six table-top objects)	PointNet baseline (3D point cloud segmentation).

Literature Review

Paper	Year and Venue	Key Points	Gaps	Dataset	Model
Subspace Prototype Guidance for Mitigating Class Imbalance in Point Cloud Semantic Segmentation	2024 / Springer	- Proposes SPG: dual-branch prototype guidance using separate feature subspaces to boost minority discrimination. - Reports SOTA on S3DIS, ScanNet v2, ScanNet200, Toronto-3D.	- Oversampling / weighted-loss methods can distort distributions and give limited gains. - Need approaches that improve balance without extra inference complexity.	S3DIS; ScanNet v2; ScanNet200; Toronto-3D	PointNet++, PTv1, PTv2.

Literature Review

Paper	Year and Venue	Key Points	Gaps	Dataset	Model
Class-imbalanced semi-supervised learning for large-scale point cloud semantic segmentation via decoupling optimization	2024 / Science Direct	- Decoupling Optimization: separate backbone & classifier training to reduce bias. - Two-Round Pseudo-Label Generation for balanced head/tail learning. - Multi-Class Imbalanced Focus Loss (MI-FL) to emphasize minority classes.	- Prior pseudo-labeling favors head classes; joint optimization causes classifier bias. - Need dynamic / adaptive loss functions for real-world settings.	S3DIS; ScanNet-V2; Semantic3D; Semantic KITTI	RangeNet++, KPRNet, PointRend, STDC, SegFormer; integrated with Sub-classified Loss / MI-FL.

Existing Approaches in 3D Segmentation

To address point cloud segmentation, current research generally follows three trajectories:

1. **Foundation Model & SAM-based Approaches**
   - Utilizes 2D-to-3D lifting techniques where Segment Anything Model (SAM) provides zero-shot masks for 3D refinement.
   - Focuses on cross-modal knowledge transfer to achieve high-level semantic priors.
2. **Geometry-based Approaches**
   - Relies on Farthest Point Sampling (FPS) and K-nearest neighbor (k-NN) grouping to extract local geometric features.
   - Uses hierarchical abstraction (PointNet++ style) to capture multi-scale spatial context.
3. **Neural Network (NN) based Architectures**
   - Employs Point-Wise MLP encoders, KPConv, or DGCNN to learn dense point features $F_{in} \in \mathbb{R}^{N \times C}$
   - Focuses on enhancing global context via feature propagation (FP) layers.

Critical Analysis of Research Gaps

Despite advancements, existing methods exhibit systemic inefficiencies when handling skewed data:

Gaps in SAM/Foundation Approaches

- **Contextual Loss:** The transition from 2D priors to 3D refinement often results in a loss of fine-grained geometric detail, especially for rare classes
- **Agnostic Prompting:** Standard SAM-based masks are not optimized for spatial confusion between adjacent minority/majority parts

Gaps in Geometry-based Approaches

- **Class-Agnostic Sampling:** FPS and density-based upsampling treat all points equally, leading to the oversampling of common structures at the expense of rare regions.
- **Density Neglect:** Standard methods ignore local neighborhood density variations (G_{conf}) critical for detecting sparse minority clusters

Gaps in NN-based Architectures

- **Feature Space Dominance:** Learned embeddings F_{out} are dominated by frequent category signatures, causing "Inertial Learning" that suppresses rare class signals.
- **Biased Boundaries:** Lack of dedicated risk distillation leads to decision boundaries that subsume minority points into majority clusters

Problem Statement

Given: A 3D point-cloud dataset $\mathcal{D} = \{X_1, \ldots, X_N\}$, where each point cloud $X_i = \{(p_j, c_j)\}_{j=1}^{n_i}$ contains coordinates $p_j \in \mathbb{R}^3$ and semantic class labels $c_j \in \{1, \ldots, K\}$, with **highly imbalanced** class distributions.

Task: Develop a model M_ϕ that improves segmentation performance for **minority classes** while preserving geometric fidelity and maintaining low computational overhead.

Objective: Solve:

$$\max_{\phi} \ \mathrm{mIoU}(M_\phi; \mathcal{D}),$$

such that the model reduces bias toward majority classes and satisfies:

$\mathrm{mAcc}(M_\phi)$ and minority-class IoU are maximized under fixed compute constr

X_i: point cloud; p_j: 3D coordinates; c_j: class label; K: number of semantic classes; mIoU: mean Intersection over Union; mAcc: mean class accuracy; ϕ: model parameters.

Core Challenge: Learn strong, unbiased features for minority classes in the presence of severe class imbalance and non-uniform point density.

Research Objectives

Objective 1: Multi-Scale Geometric Feature Extraction
To develop a hierarchical preprocessing and encoding pipeline using Farthest Point Sampling (FPS) and PointNet-inspired set abstraction to capture both local geometric signatures and global context across $N \to 1$ point-count scales

Objective 2: Interactive Prompt Refinement and Conflict Resolution
To design a Negative Point Contrast (NPC) mechanism that identifies spatial confusion between dense 3D parts via IoU-based conflict detection and utilizes subtractive prompting to sharpen part-level decision boundaries

Objective 3: Uncertainty-Guided Minority Class Distillation
To strengthen minority-class representation by inverting μ-Relief logic to calculate a Geometric Confusion Score (G_{conf}), thereby transforming geometric uncertainty into a supervisory risk signal (H_{score}) for "hard" point mining

Objective 4: Global Semantic Regularization for Balanced Learning
To implement a three-headed architecture (NPC, Risk, PBR) that utilizes Hungarian Matching to align batch-level features with a global prototype buffer, ensuring semantic consistency across skewed training distributions

Architecture Overview

Core Concept: A hierarchical point cloud segmentation framework integrating prompt-based refinement and uncertainty-aware mining.

- **Backbone:** PointNet-inspired hierarchical encoder.
  - Utilizes Farthest Point Sampling (FPS) and K-nearest neighbor grouping.
  - Hierarchical abstraction ($N \to 512 \to 128 \to 1$) to extract global context.
- **Multi-Head Design:**
  1. **Head 1: NPC Refinement:** Handles spatial confusion via subtractive prompting.
  2. **Head 2: Prototype Based Regularization (PBR):** Aligns features with global semantic categories.
  3. **Head 3: Geometric Risk Head:** Distills geometric complexity using Inverted μ-Relief logic.

Architecture Overview

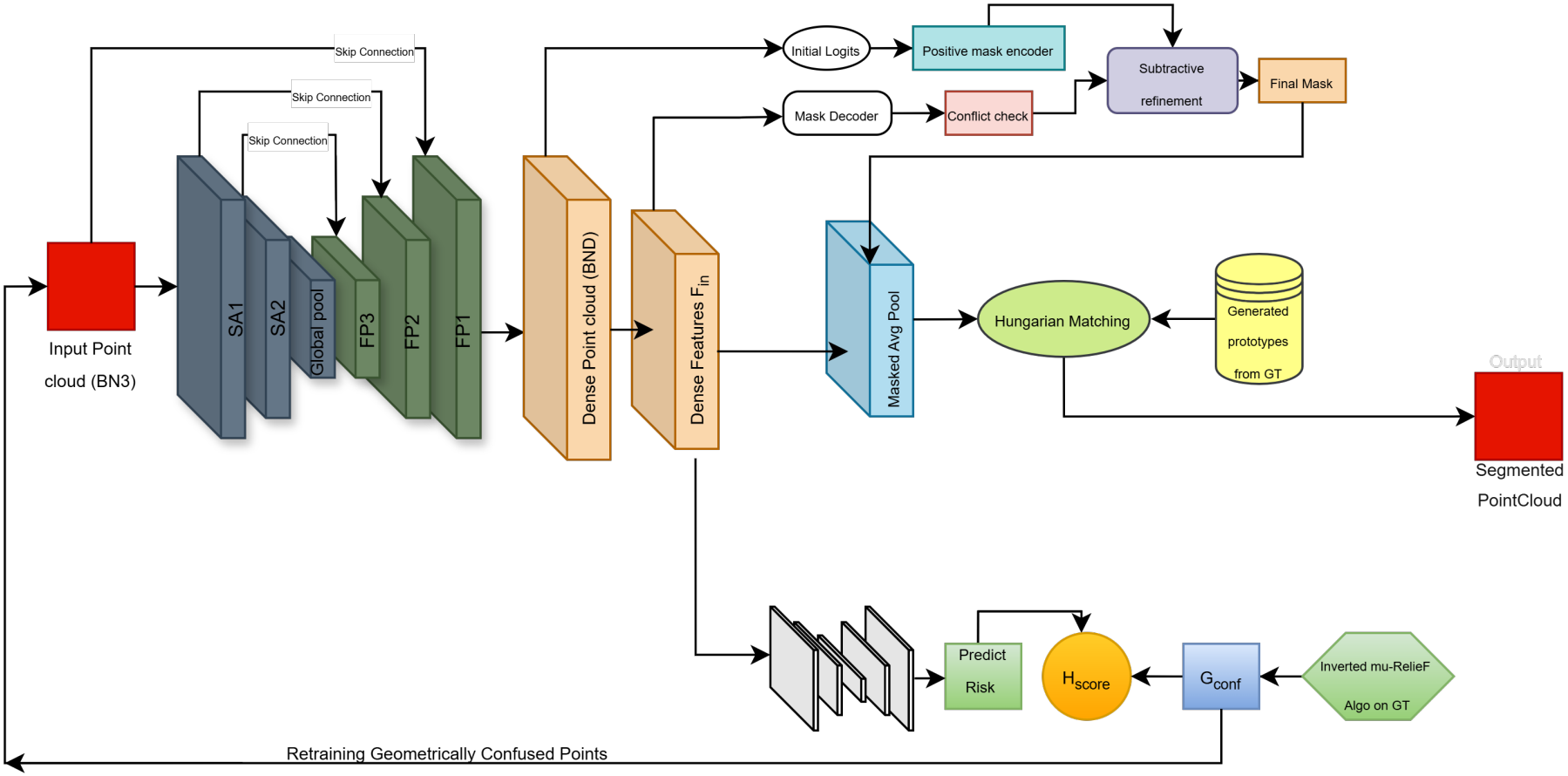


Figure: Architecture Overview of the proposed 3 Headed Model

Architecture Overview

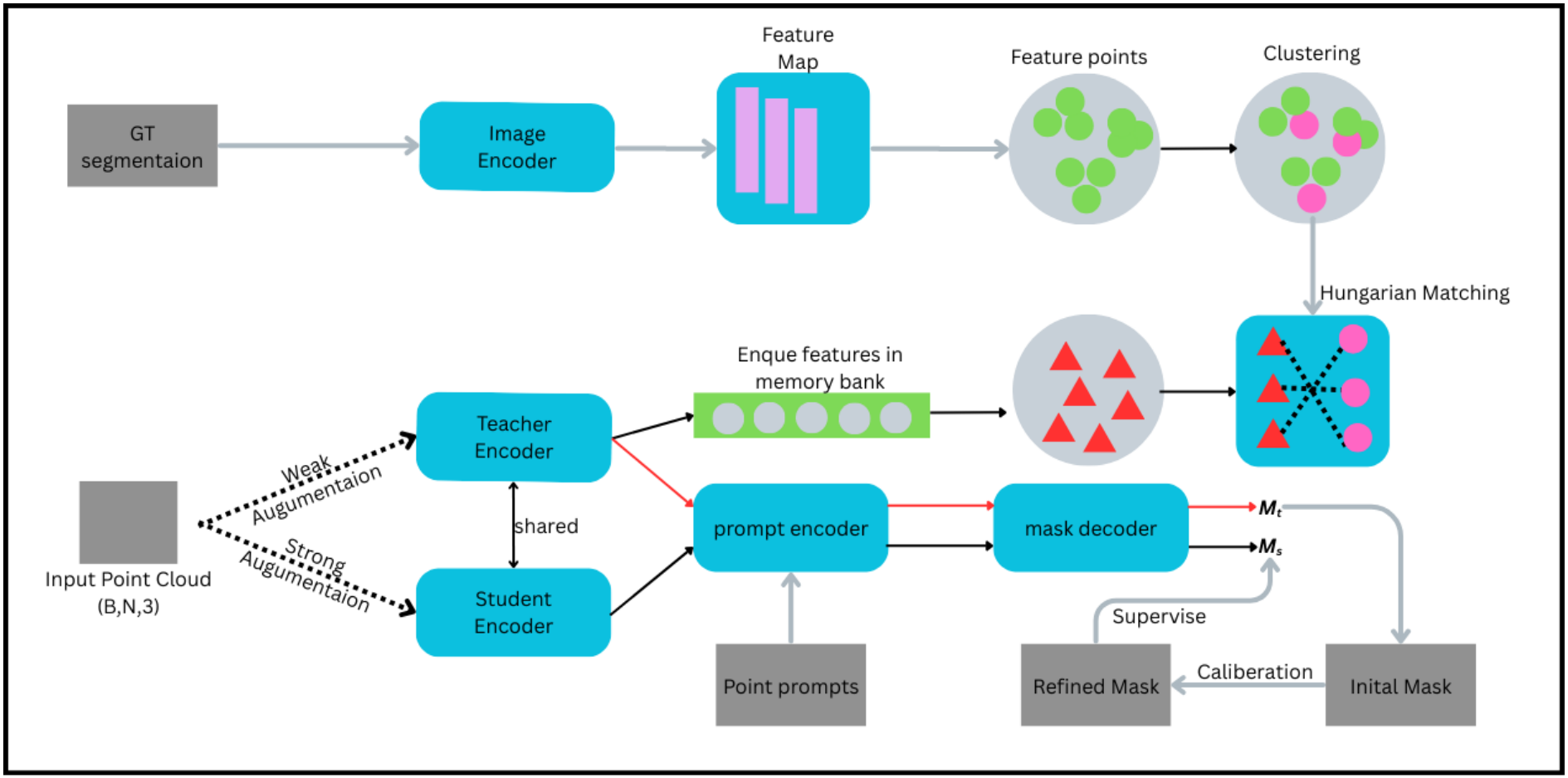


Figure: Architecture Overview of the base model: PointSAM

Head 1: Negative Point Contrast (NPC) Refinement

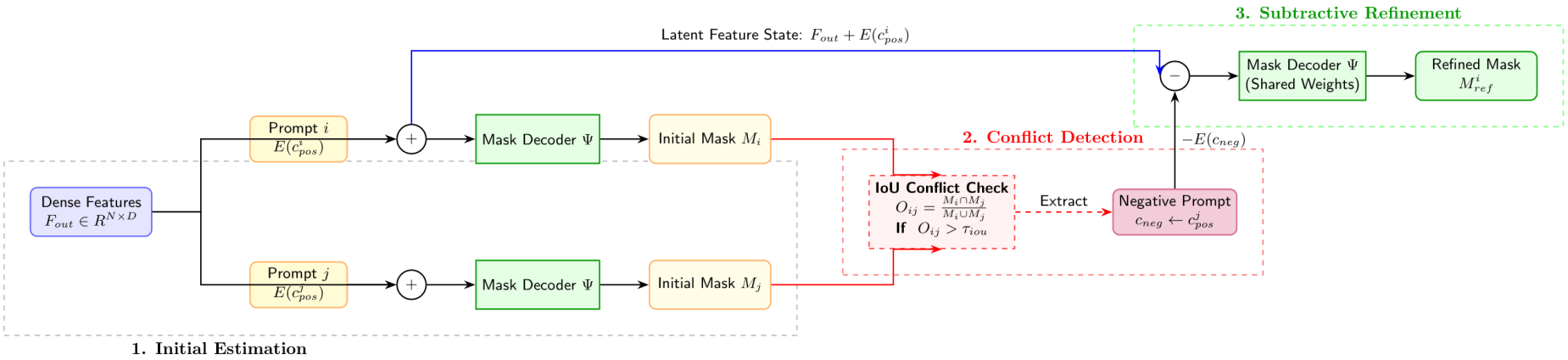


Figure: Graphical Representation of NPC Head

Head 1: Negative Point Contrast (NPC) Refinement

Objective: Resolve spatial confusion between adjacent parts using subtractive prompting

Algorithm Phase:

1. **Initial Estimation:** Generate masks using positive prompts c_{pos} combined with dense features F_{out}.

$$L_{init} = \Psi_{decoder}(F_{out} + E(c_{pos}))$$

2. **Conflict Detection:** Compute IoU matrix O_{ij}. If $O_{ij} > \tau_{iou}$ (0.1), parts are spatially confused.
3. **Subtractive Prompting:** Inject the conflicting coordinate as a Negative Prompt (c_{neg}) to push decision boundaries away from confusion.

$$L_{refined} = \Psi_{decoder}(F_{out} + E(c_{pos}) - E(c_{neg}))$$

NPC Refinement: PointSAM vs. H3-PSNet

Feature	Original PointSAM	H3-PSNet (Ours)
Trigger	User-defined or iterative error detection.	**Automatic IoU overlap** detection ($IoU > 0.1$).
Selection	Random selection within error regions.	**Cross-Part Distractor**: Active selection from competing masks.
Operation	Embedding concatenation in token sequence.	**Subtractive Embedding**: Feature suppression via $F + P_{pos} - P_{neg}$.
Objective	Edge refinement in 2D imagery.	**Spatial Disentanglement** of adjacent 3D parts.

Head 2: Prototype Based Regularization (PBR)

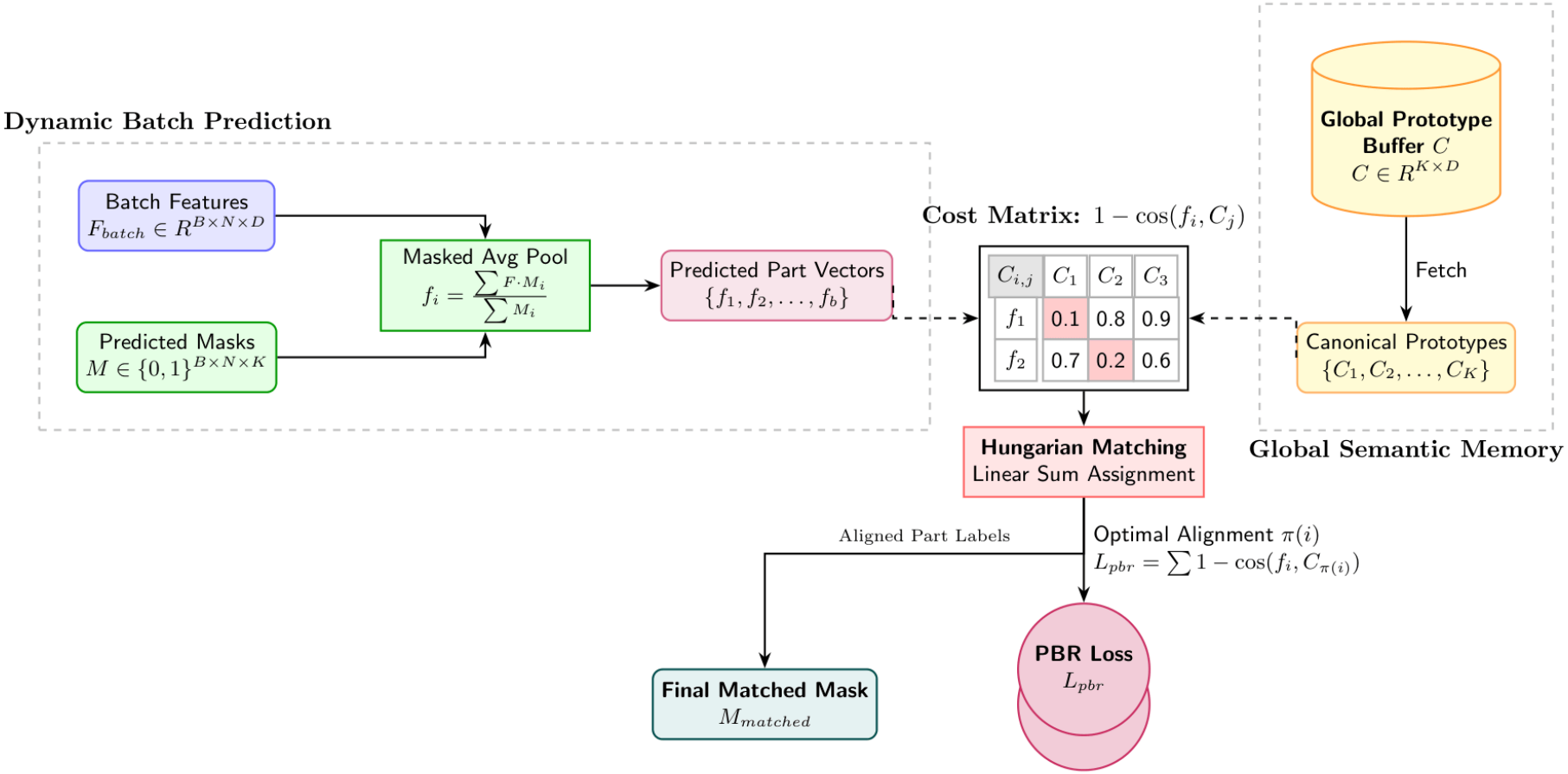


Figure: Depiction of PBR Head

Head 2: Prototype Based Regularization (PBR)

Objective: Ensure learned features align with global semantic clusters (e.g., all "Chair legs" cluster together).

Mechanism:

- Maintain a global prototype buffer $C \in \mathbb{R}^{K \times D}$.
- Compute Hungarian Matching cost between current batch features f_i and global prototypes using Cosine Distance.

Loss Formulation:

$$L_{pbr} = \min_{\pi \in \Pi} \sum_{i=1}^{B} (1 - \cos(f_i, C_{\pi(i)})) \tag{1}$$

Where f_i is the average feature vector of the i-th predicted part.

Key Benefit: Stabilizes training by enforcing global consistency across batches.

PBR Regularization: PointSAM vs. H3-PSNet

Feature	Original PointSAM	H3-PSNet (Ours)
Philosophy	Teacher-Student pseudo-mask distillation.	**Global Memory**: Alignment with pre-computed "ideal" prototypes.
Loss Function	standard Cross-Entropy / Dice Loss.	**Hungarian Matching**: Bipartite matching between batch and global buffer.
Feature Space	Instance-specific optimization.	**Category-Invariant**: Cross-instance semantic clustering.
Memory	No cross-batch feature storage.	**Global Buffer (C)**: Maintains persistent part centroids.

Head 3: Geometric Risk Distillation

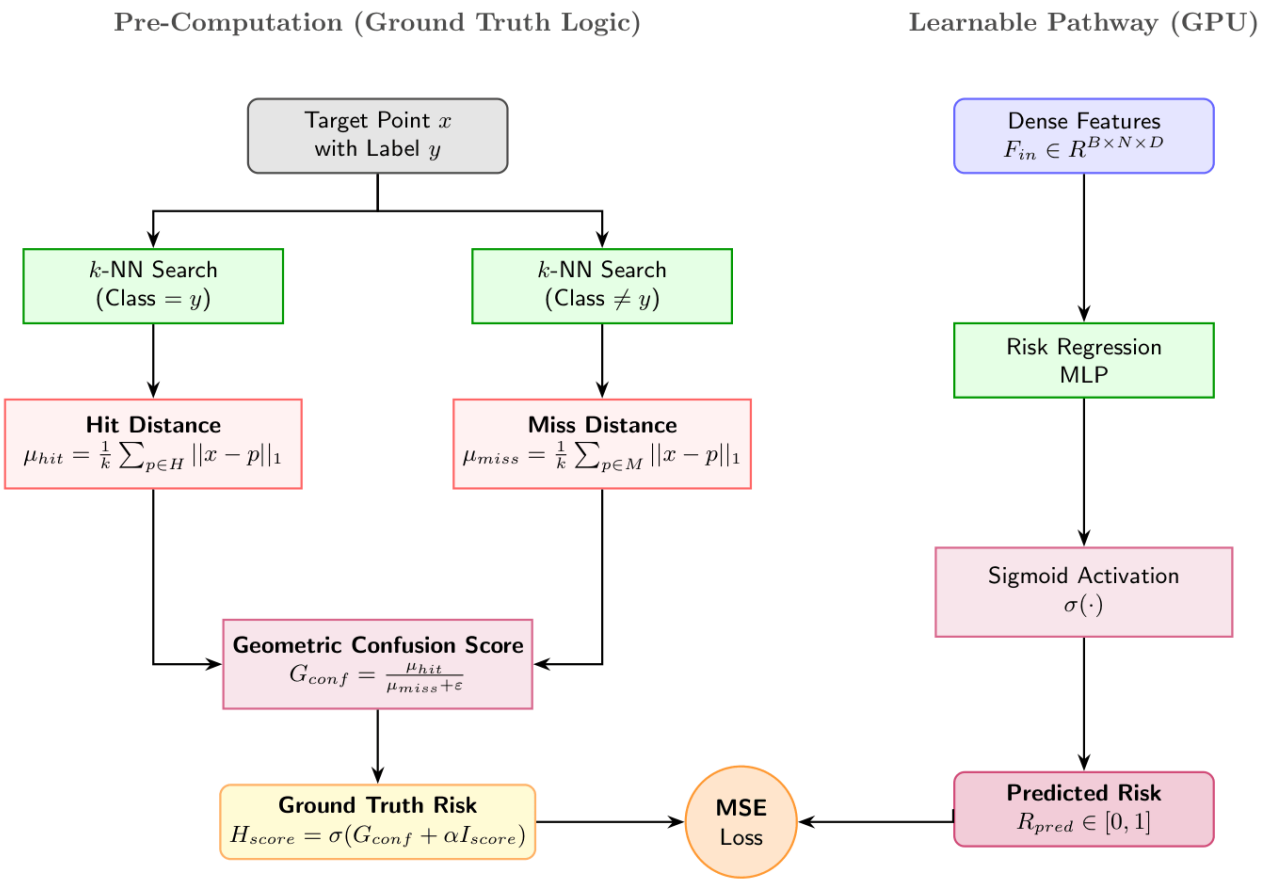


Figure: Graphic depiction of Inverted μ-Head for Risk detection

Head 3: Geometric Risk Distillation

Objective: Mine for "hard points" (boundary confusion) using Inverted μ-Relief.

Mathematical Formulation:

- **Hit Distance (μ_{hit}):** Avg distance to k-nearest neighbors of the *same* class.
- **Miss Distance (μ_{miss}):** Avg distance to k-nearest neighbors of *different* classes.

Geometric Confusion Score (G_{conf}):

$$G_{conf}(p_i) = \frac{\mu_{hit}}{\mu_{miss} + \varepsilon} \tag{2}$$

Interpretation: $G_{conf} \approx 1$ implies the point is hard to classify (boundary point).

Final Risk Score (H_{score}): Combines geometry with class rarity (I_{score}):

$$H_{score} = \sigma(G_{conf} + \alpha \cdot I_{score}) \tag{3}$$

Where $I_{score} = \log(N_{total}/N_c)$ handles class imbalance.

Training Strategy: Blended Mining

Cyclic Learning Strategy: We propose a cyclic loop ($T = 30$) switching between base training and hard-mining phases.

Algorithm: Blended Mining Loop

Input: Cycle $T = 30$, Base Epochs $E_{base} = 20$

If $(epoch \mod T) > E_{base}$: *// Mining Phase*
  Calculate Risk R_{gt} via Relief Score
  $M_{mask} \leftarrow (R_{gt} > \tau_{hard})$ *// Keep only hard points*
Else:
  $M_{mask} \leftarrow 1$ *// Keep all points*
End If

Forward Pass:
  $L_{seg}, F, R_{pred} \leftarrow Model(P)$
Weighted Loss:
  $L_{seg} \leftarrow BCE(L_{seg}, Y) \cdot (1 + R_{gt}) \cdot M_{mask}$

Total Loss Function

The model is optimized end-to-end using a multi-task objective function:

Final Loss Equation

$$L_{total} = L_{seg} + \lambda_1 L_{risk}(R_{pred}, R_{gt}) + \lambda_2 L_{pbr}(F, C)$$

Hyperparameter Specifications:

- **Ranges:** Typically, $\lambda_1 \in [0.1, 1.0]$ and $\lambda_2 \in [0.01, 0.1]$.
- **Geometric Risk Weight (λ_1):** Governs the distillation of H_{score} relative to segmentation.
- **Regularization Weight (λ_2):** Controls the Hungarian Matching distance for global prototypes.

Correlation and Balancing:

- **Inverse Scaling:** As λ_1 increases during mining phases to emphasize "hard" points, λ_2 must remain stable to prevent feature drift from the global prototypes C.
- **Synergy:** L_{risk} identifies boundary uncertainty, while L_{pbr} ensures that even under high risk, the part features f_i remain semantic-aligned.

Implementation Details: Hyperparameters

Key System Parameters: The following thresholds and constants are utilized to govern the blended mining and prompt refinement processes.

Parameter	Value	Description
τ_{iou}	0.1	Spatial confusion threshold (NPC)
T	30	Blended mining cycle length
E_{base}	20	Base training epochs per cycle
E_{mine}	10	Mining epochs per cycle
K	16	Neighbors for μ-Relief selection
α	0.25	Class imbalance weight

Table: Core implementation constants.

- **Feature Distance:** All geometric calculations utilize the Manhattan (L_1) distance.
- **Confusion Buffer:** A small constant ϵ is used to prevent division by zero in the confusion ratio.

Implementation Details: Technical Execution

Backbone and Encoding:

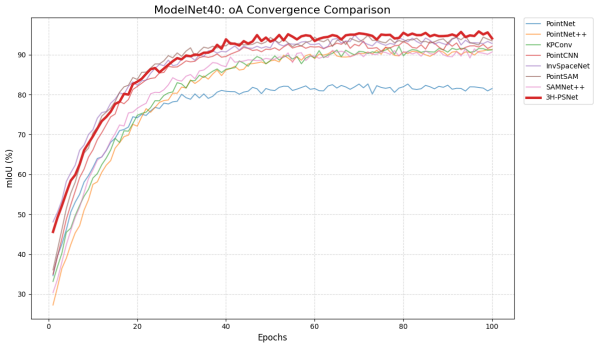
- **Sampling Strategy:** Farthest Point Sampling (FPS) is used to select a subset of centroids $\{x_j\}_{j=1}^{M}$.
- **PointNet Abstraction:** For each centroid, K neighbors are grouped within radius r for feature extraction
- **Data Inversion:** The Geometric Risk head calculates G_{conf} to identify outliers where $\mu_{hit} > \mu_{miss}$.

Optimization Environment:

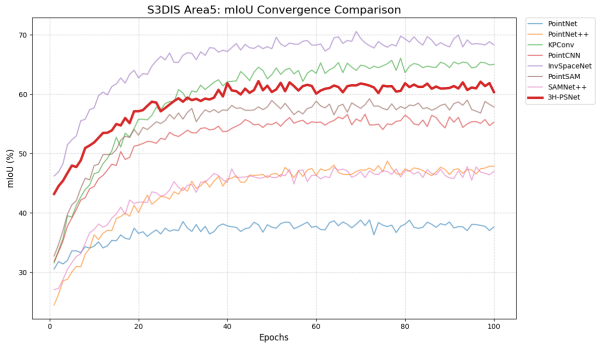
- **Global Consistency:** A prototype buffer C is maintained to cluster global semantic categories.
- **Prompting:** Part segmentation is treated as a conditional prompting task using positive coordinate embeddings.
- **Risk Weighting:** Segmentation loss L_{seg} is dynamically weighted by $(1 + R_{gt})$ during training.

Convergence Analysis

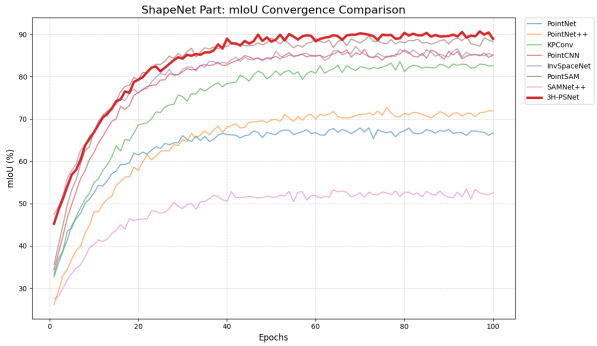
Training Dynamics: Comparison of mIoU convergence across datasets. Our method (red) demonstrates faster stability and higher peak performance.



(a) ModelNet40 (oA)



(b) S3DIS Area5 (mIoU)



(c) ShapeNet Part (mIoU)

- **ModelNet40:** Rapid convergence reaching $\approx 95\%$ accuracy.
- **S3DIS:** Significant gap maintained over PointSAM and Base models.
- **ShapeNet:** Consistent dominance in part segmentation tasks.

Quantitative Comparison on Standard Benchmarks

Performance Summary: Comparison with SOTA methods.

2*Model	ModelNet40			S3DIS Area5			ShapeNet Part		
	oA	mAcc	mIoU	oA	mAcc	mIoU	oA	mAcc	mIoU
PointNet	81.77	-	-	75.16	47.20	37.87	85.34	77.01	66.98
PointNet++	90.70	87.40	-	78.74	47.19	47.24	81.65	78.18	71.33
KPConv	91.23	85.50	-	**84.34**	68.58	**64.98**	80.93	79.12	82.62
PointCNN	92.20	88.10	-	81.28	50.09	55.34	94.98	90.08	85.21
InvSpaceNet	93.20	90.10	-	86.10	**74.00**	68.60	87.20	84.80	85.50
PointSAM	93.50	-	-	76.56	-	57.87	**95.89**	**90.65**	88.28
SAMNet++	90.45	-	-	80.74	56.76	46.74	93.29	88.77	52.37
3H-PSNet (Ours)	**94.98**	-	-	83.58	-	61.24	95.34	90.27	**89.85**

Table: Comparison of Overall Accuracy (oA), Mean Accuracy (mAcc), and Mean IoU (mIoU).

**Note: Our 3-Headed model achieves competitive or superior performance across all benchmarks, particularly in ShapeNet Part segmentation.*

ShapeNet Part: Class-wise Performance

Detailed Analysis: Comparison of mIoU across 16 object categories.

Class	PointSAM	Ours	SAMNet++
Airplane	88.96	**89.03**	40.18
Bag	31.96	**68.47**	49.66
Cap	36.54	**70.52**	64.08
Car	88.40	**94.32**	39.40
Chair	80.48	**95.31**	30.53
Earphone	69.24	**78.90**	44.72
Guitar	90.12	**93.93**	62.78
Knife	69.80	69.45	**77.63**

Class	PointSAM	Ours	SAMNet++
Lamp	86.83	**89.09**	40.58
Laptop	91.64	**91.86**	88.16
Motorbike	82.52	**95.04**	22.19
Mug	69.77	**93.72**	77.73
Pistol	68.88	**76.97**	52.71
Rocket	47.84	**48.35**	30.15
Skateboard	79.09	**80.35**	56.01
Table	**81.51**	76.55	52.12

Key Observation: Our model significantly outperforms baselines in geometrically complex classes like *Bag*, *Cap*, and *Mug*.

Conclusion

Summary of Contributions:

- **Novel Mining Mechanism:** Proposed Inverted μ-Relief to systematically identify and mine geometrically ambiguous "hard points."
- **Geometry as Supervision:** Transformed local geometric confusion into a robust learning signal, effectively sharpening decision boundaries.
- **Performance Gains:** Demonstrated superior convergence stability and significant accuracy improvements on minority classes.
- **Principled Efficiency:** Establishes a data-efficient alternative to brute-force reweighting by prioritizing uncertainty-guided refinement.
- **Broader Impact:** Bridges the gap between foundation model ideas and imbalance-aware 3D learning.

Future Directions

We identify three key avenues to extend this research:

- **Semi-Supervised Expansion:** Leverage the proposed risk scores as confidence metrics for pseudo-labeling in label-scarce environments.
- **Adaptive Density Handling:** Investigate dynamic neighborhood sizing (k-NN) to handle varying point densities in outdoor scenes.
- **Real-Time LiDAR Deployment:** Scale the architecture to process raw, continuous LiDAR streams for autonomous navigation.

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